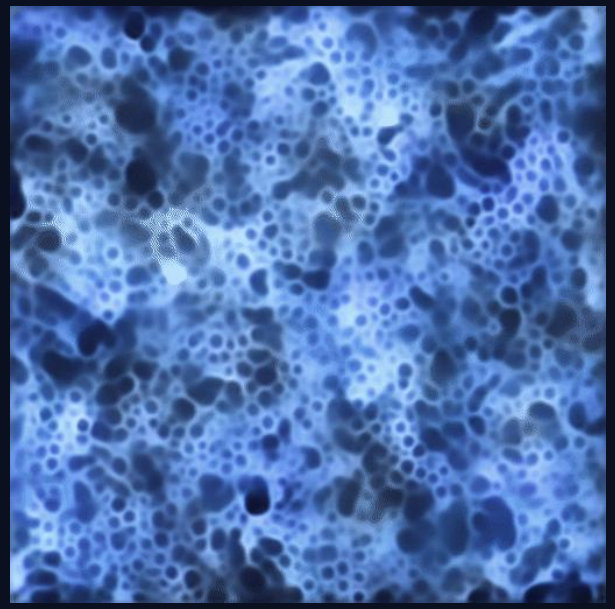
DNA to Display

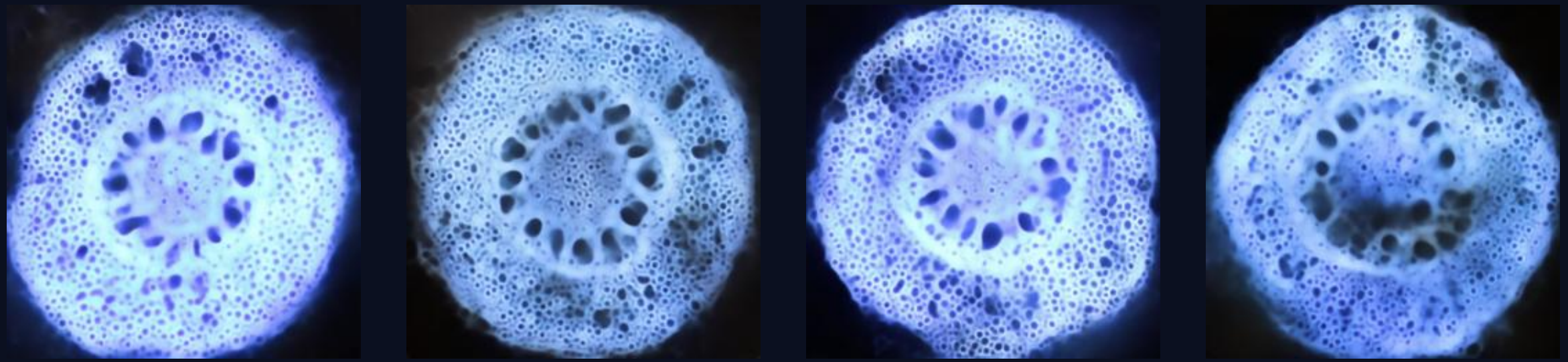
Generating phenotypic images from genetic information

Tyler Mitts
Computer Science
University of Missouri



ONE MAIZE GENOTYPE

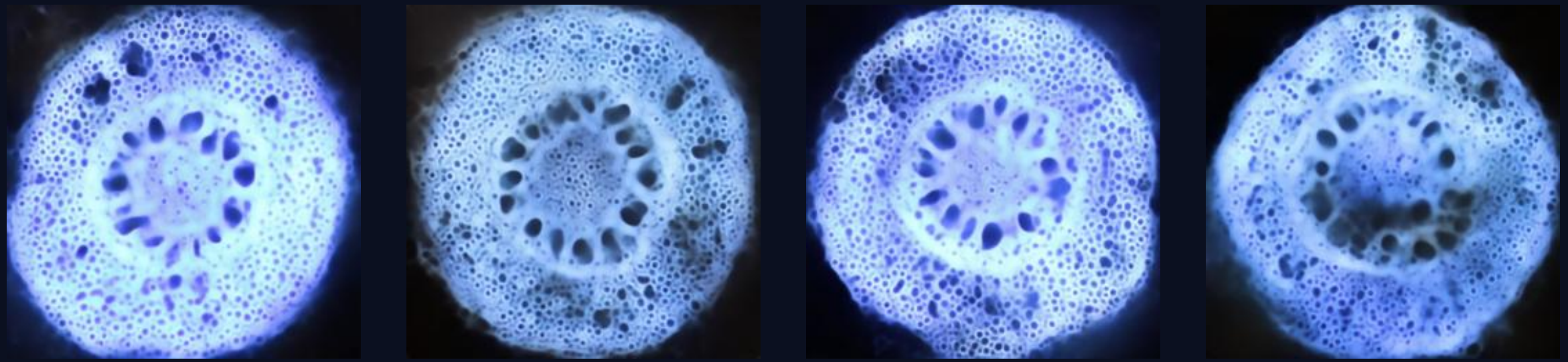
Inside a maize root: vessels, stele, cortex



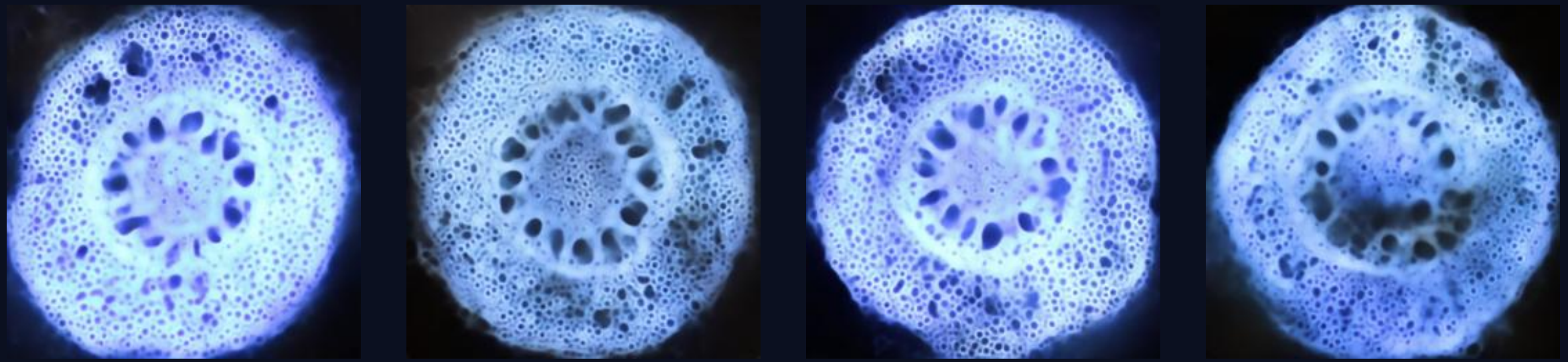
Cross-sections through four nodal roots of the same maize genotype, showing the same key features: the stele with water-carrying vessels, the cortex around it, the air spaces in between.

What if you could see inside a plant’s roots without ever growing one?

Just from a sample of its DNA.



NOT REAL



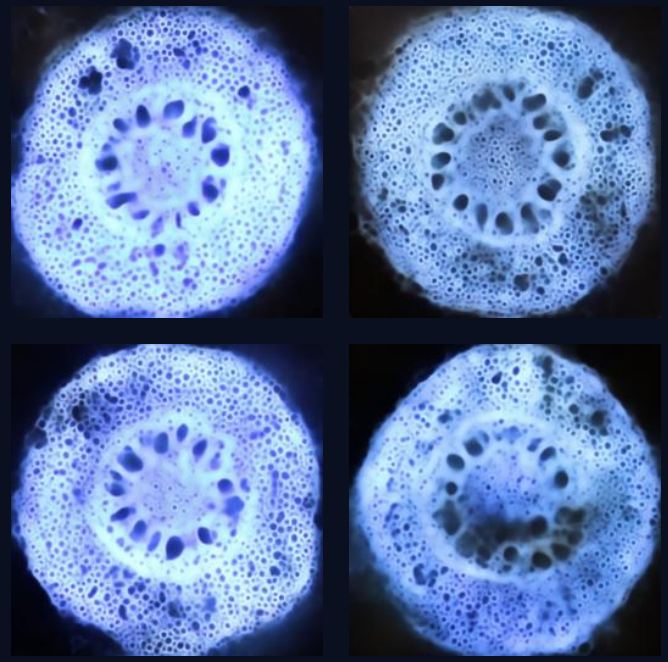
No seed was planted. No section was cut.

A computer drew all four, from this line’s DNA and nothing else.



ITS DNA, AS THE MODEL READS IT

the model

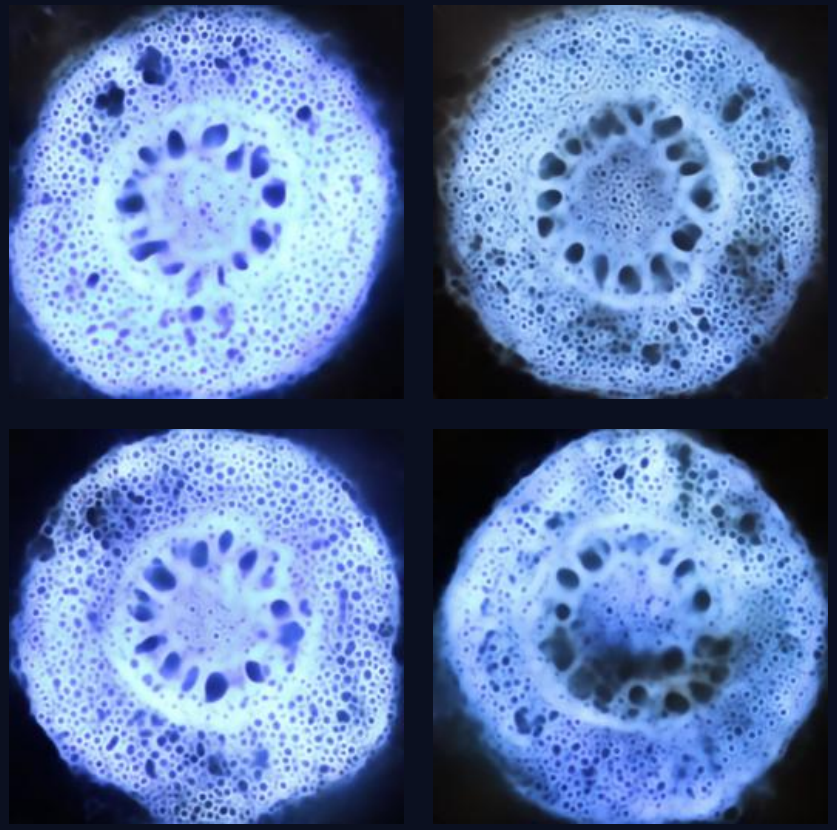


FOUR GENERATIONS, ONE GENOTYPE

MEMA042

Side by side with a real root of the same line

GENERATED



The model had never seen this genotype.

REAL



OVERVIEW

Predicting a root's anatomy from its DNA

01	Problem	Root images cost weeks of labour. Computation is cheap.
02	The data	Cross-sectional root images, and the traits we measure in them.
03	The architecture	The models that bring image generation together with genetic information.
04	Testing	How we put those models to the test.
05	Implications	The future possibilities and directions of this project.

WHY BOTHER

An image takes weeks. A genotype takes a leaf.



EVERY TRAINING IMAGE STARTS HERE

AN IMAGE

grow the plant · harvest the root · fix and section it · image the section

weeks of work, one plant at a time

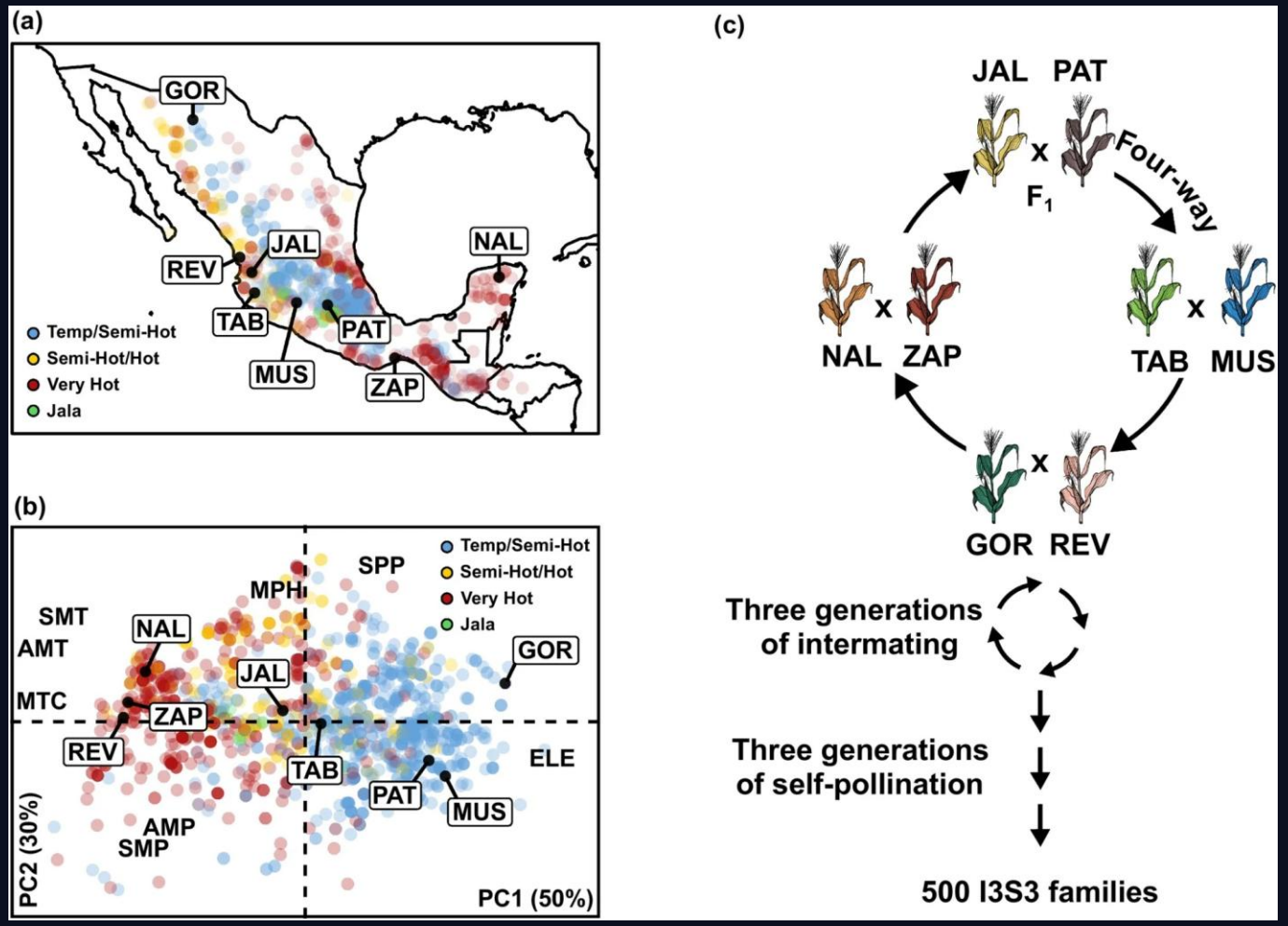
A GENOTYPE

take a leaf punch · sequence it once · reuse it forever

already done for thousands of lines

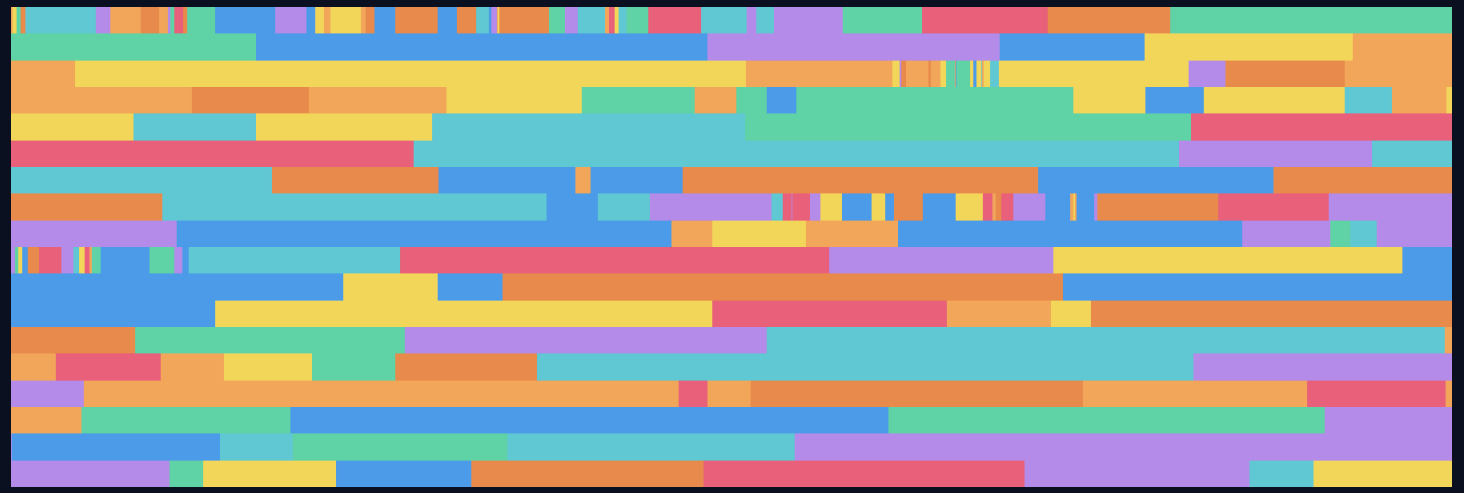
If the cheap measurement could stand in for the expensive one, anatomy could be screened at the scale genotyping already works at. That is the whole motivation, and it is not specific to roots.

MexiMAGIC: 8 Mexican parents, one population



THE DATA

Every genome is a combination of the eight parents



MEMA042 · ALL 43,788 MARKERS, COLOURED BY THE PARENT EACH CAME FROM

1 2 3 4 5 6 7 8



ONE OF ITS ROOTS

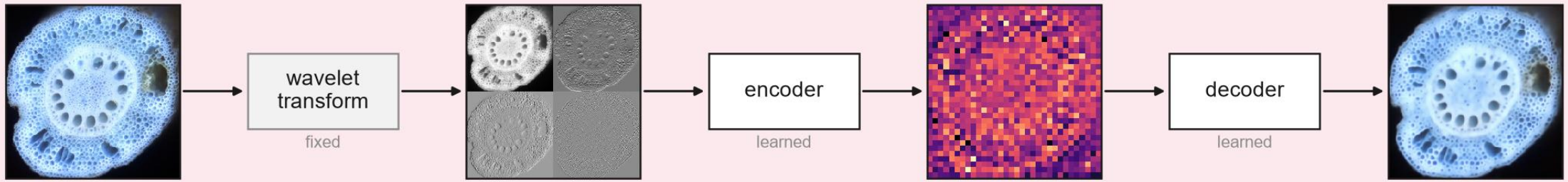
1,008
root cross-sections

200
lines with genotype

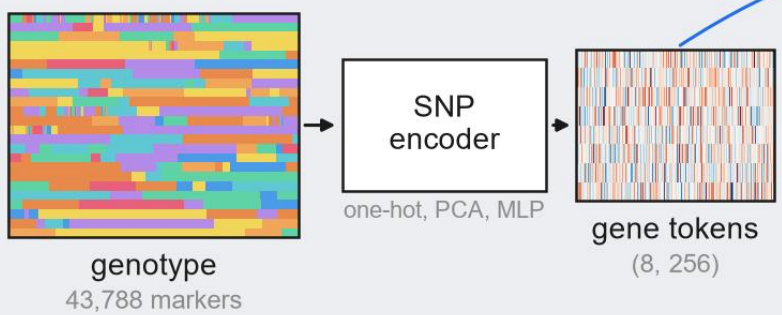
137
lines with both

27
kept out of training

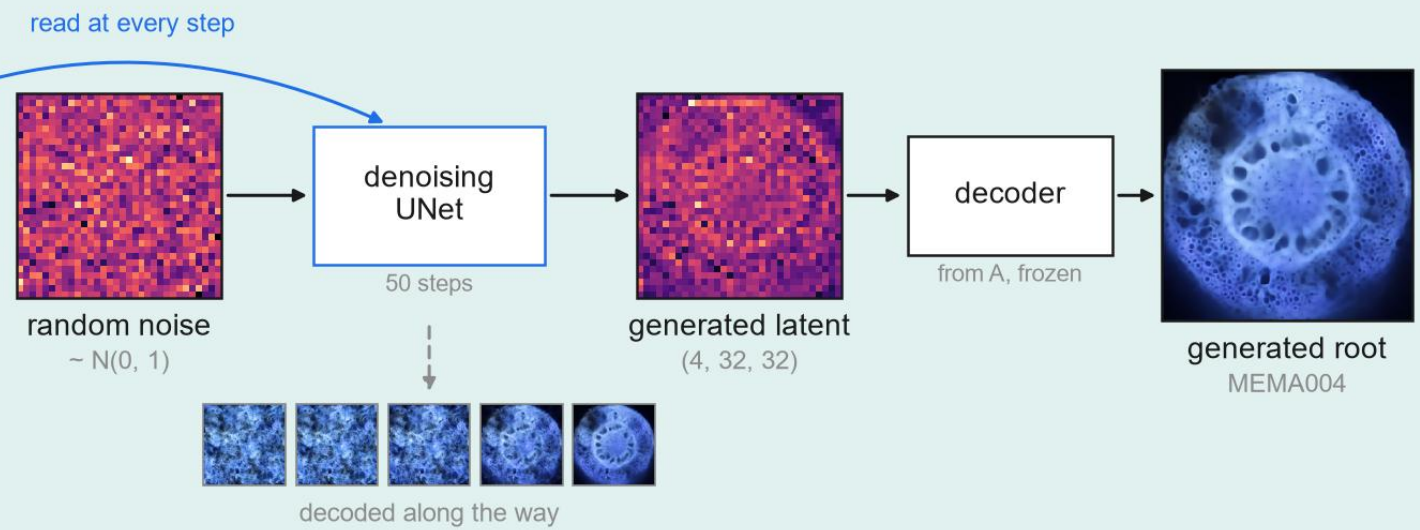
A LiteVAE: shrink a root image to a latent, and rebuild it



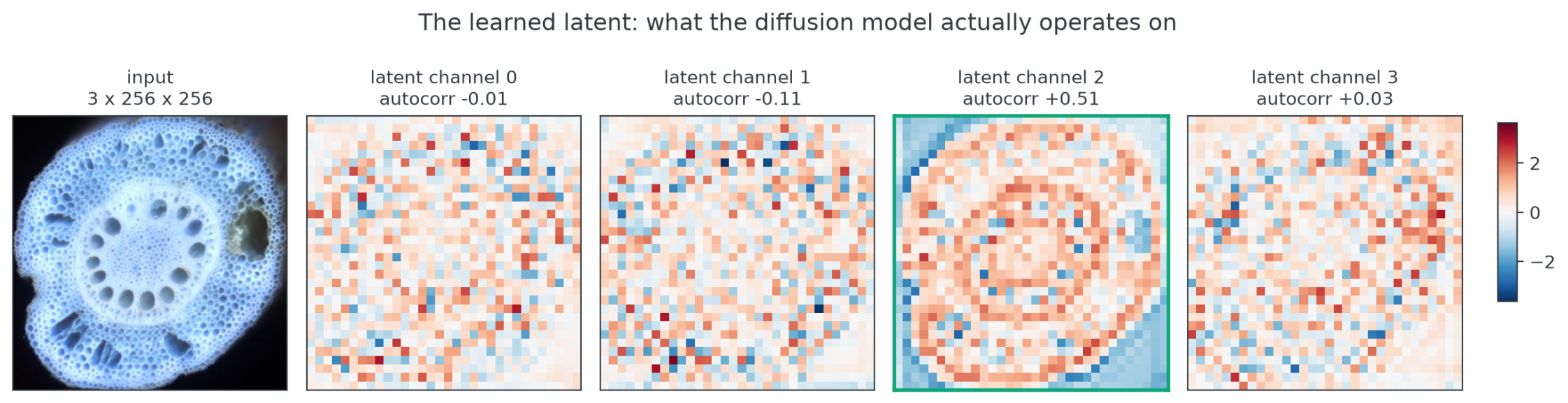
B SNP encoder: genome to tokens



C Diffusion model: noise and tokens to a new root



Step 1: LiteVAE compresses each root image 48×



IN

A root image.
3 × 256 × 256 = 196,608 numbers.

OUT

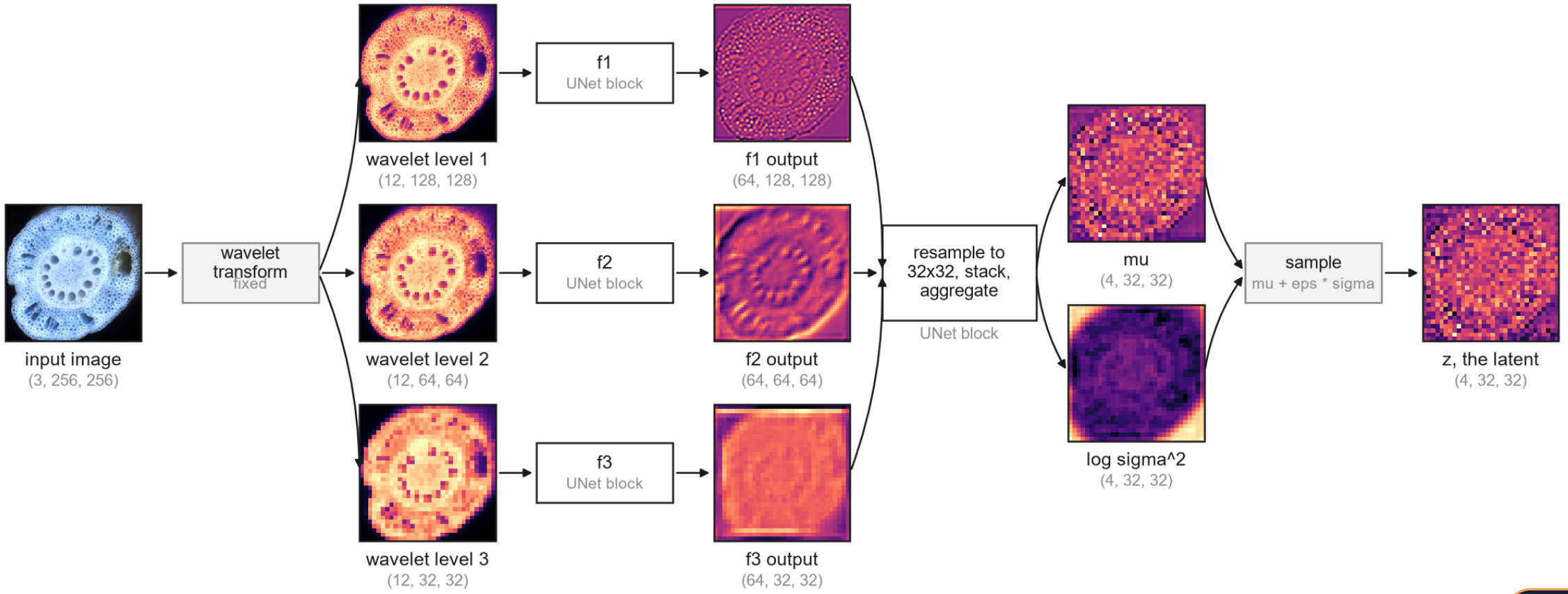
A compressed version, 4 × 32 × 32 = 4,096 numbers, and a decoder that turns it back into a picture.

WHY

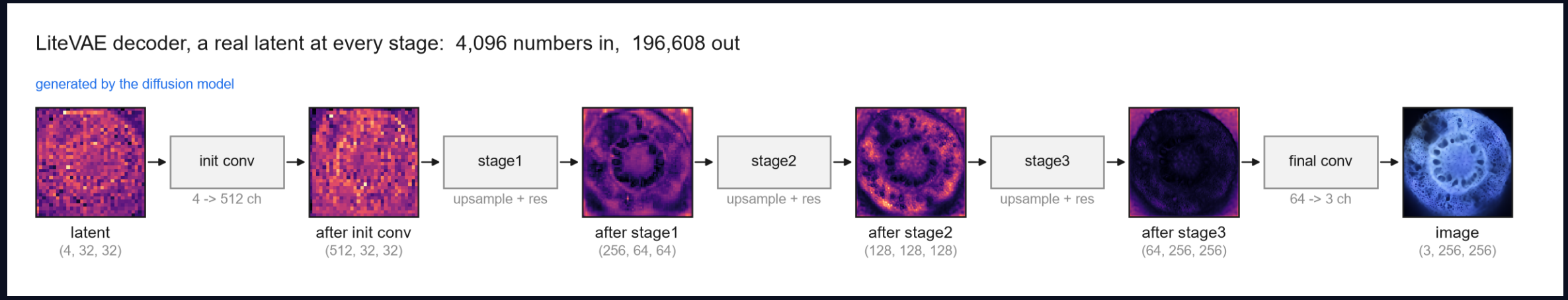
With ~700 training images, learning 196,608 numbers directly is expensive. Compressing 48× makes the problem small enough, and the traits still survive at r = 0.94–0.99.

Inside LiteVAE: a root becomes a 4×32×32 latent

LiteVAE encoder, one real root at every stage: 196,608 numbers in, 4,096 out

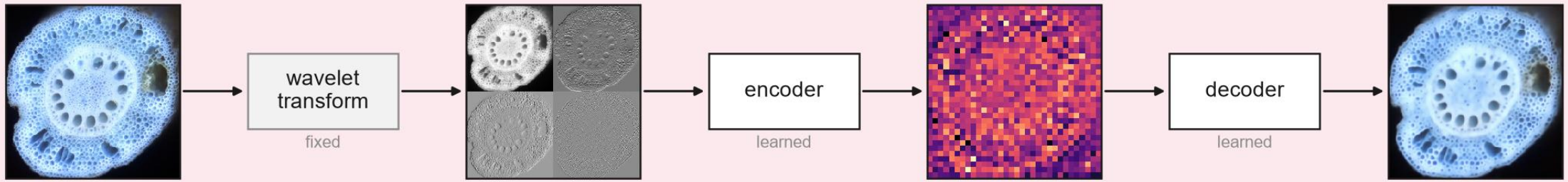


The decoder turns a latent back into a root

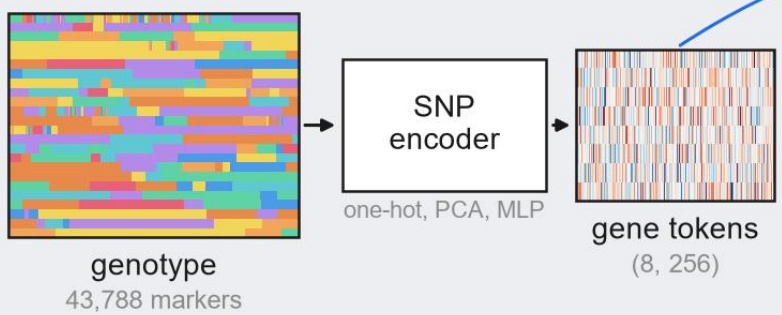


The compressor from the start of the talk, run backwards. Each stage doubles the resolution and cuts the channel count, and none of them sees the genotype. The decoder is frozen and treats a generated latent exactly as it treats an encoded one.

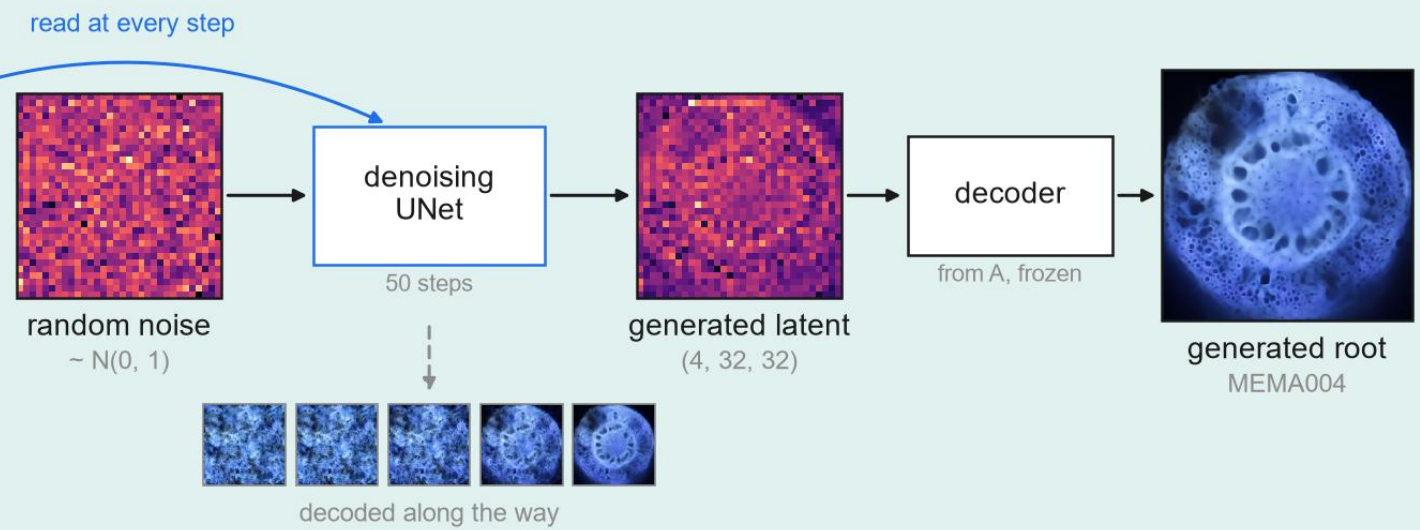
A LiteVAE: shrink a root image to a latent, and rebuild it



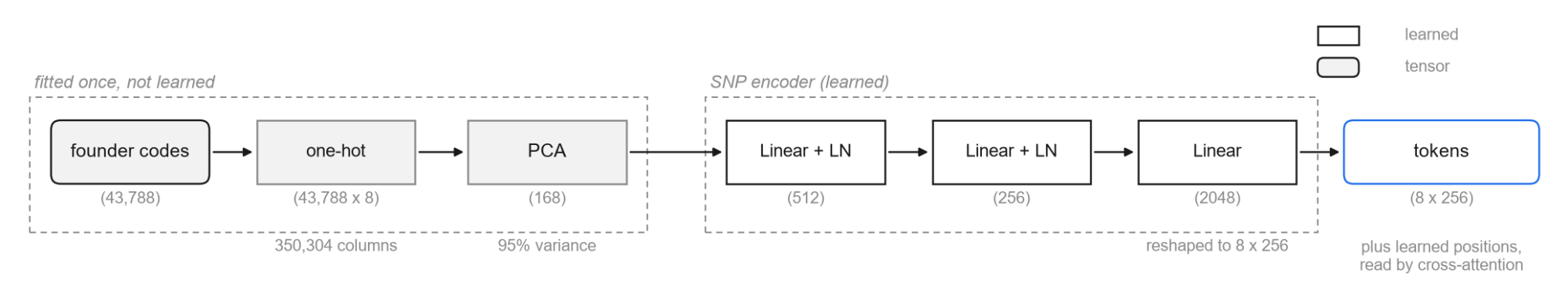
B SNP encoder: genome to tokens



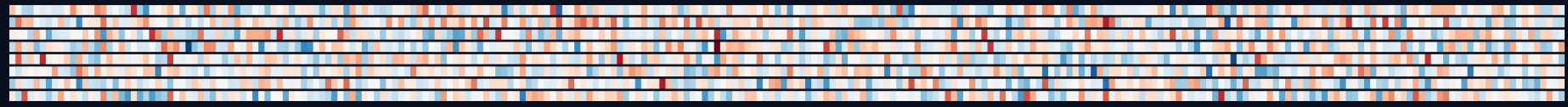
C Diffusion model: noise and tokens to a new root



Step 2: compress the genome into eight tokens



WHAT COMES OUT, FOR MEMA042 · 8 TOKENS × 256 NUMBERS · RED = POSITIVE, BLUE = NEGATIVE



IN

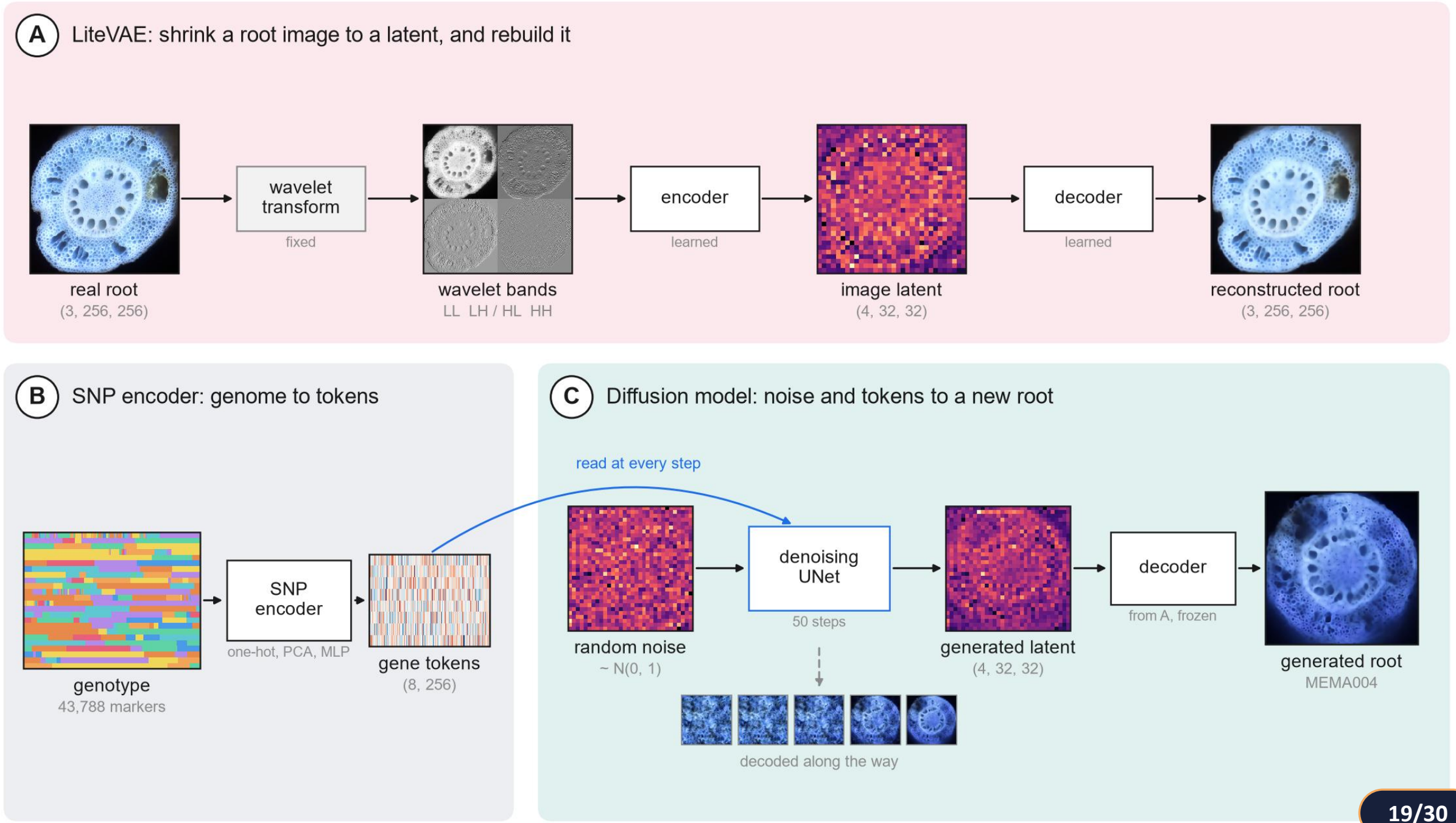
One line's genotype: 43,788 markers, each naming the founder it came from.

OUT

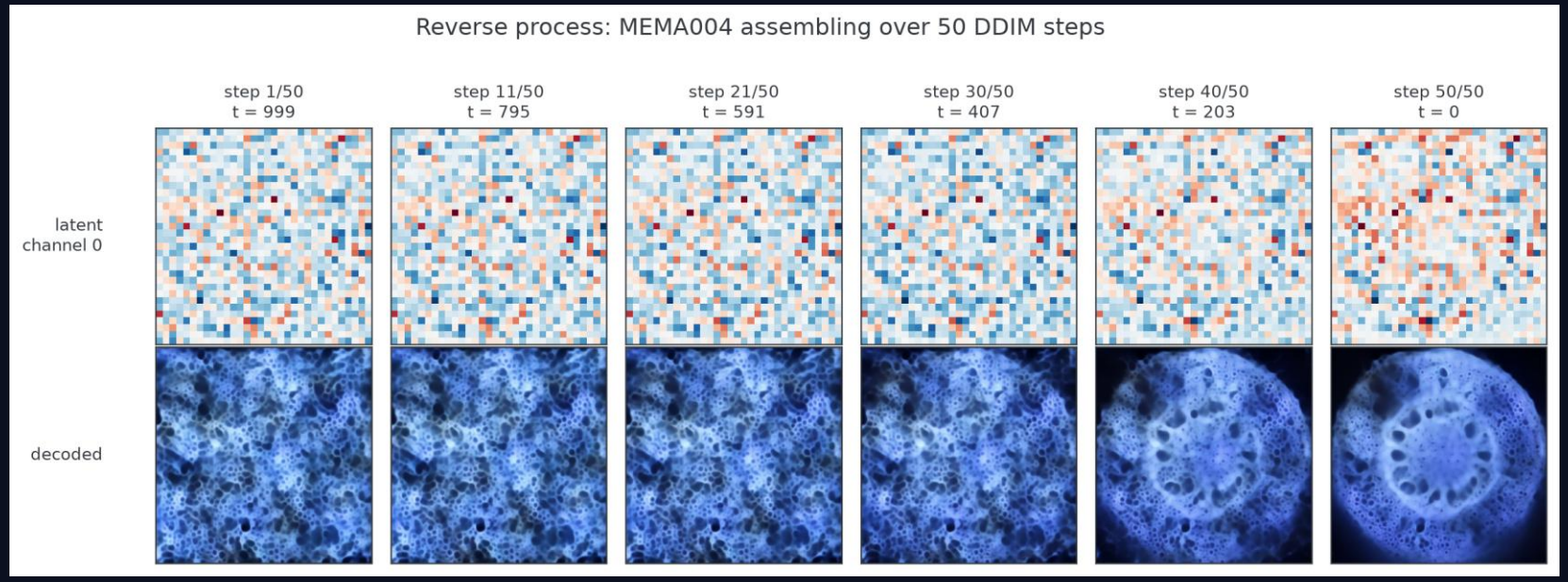
Eight tokens, which serve as short summaries the image model can read at every step.

WHY AN ENCODER AT ALL

350,000 columns against 110 lines invites memorising, and a lot markers say nothing about anatomy. Training the encoder with the generator keeps the ones that do.


A
LiteVAE: shrink a root image to a latent, and rebuild it
real root
(3, 256, 256)
wavelet transform
fixed
wavelet bands
LL LH / HL HH
encoder
learned
image latent
(4, 32, 32)
decoder
learned
reconstructed root
(3, 256, 256)
B
SNP encoder: genome to tokens
genotype
43,788 markers
SNP encoder
one-hot, PCA, MLP
gene tokens
(8, 256)
C
Diffusion model: noise and tokens to a new root
read at every step
random noise
~ N(0, 1)
denoising UNet
50 steps
generated latent
(4, 32, 32)
decoder
from A, frozen
generated root
MEMA004
decoded along the way

Step 3: turn static into a root, guided by DNA



IN

A patch of random noise, the eight genotype tokens, and how noisy the picture currently is.

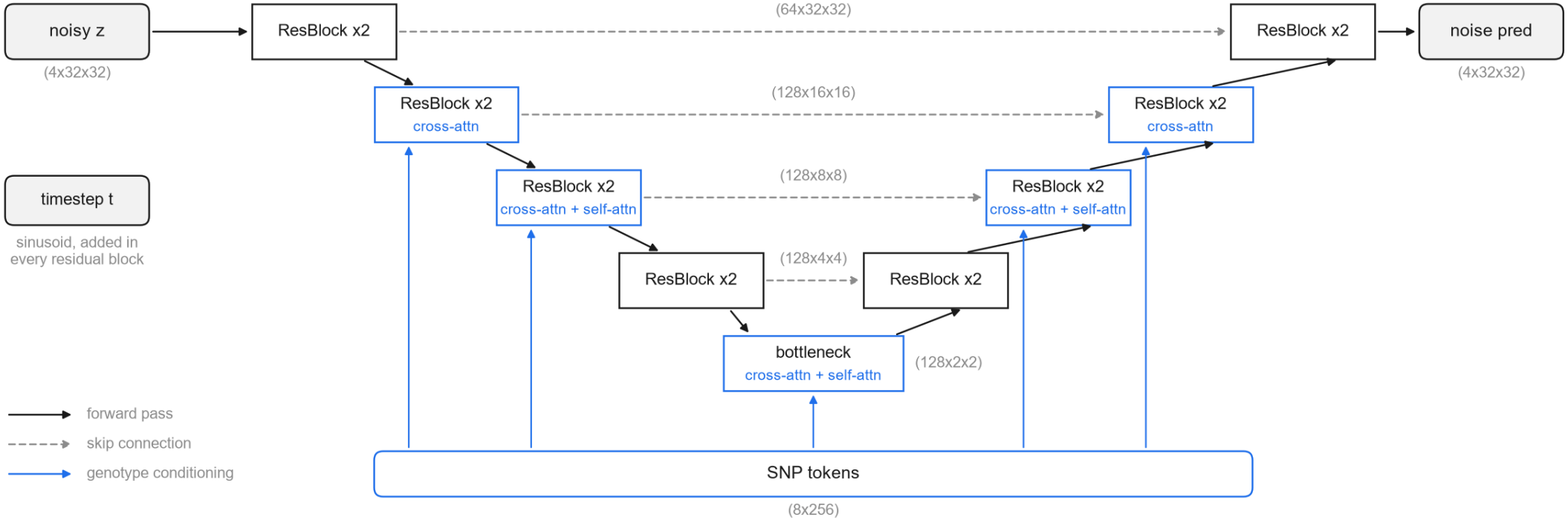
OUT

A little less noise each round. After fifty the static has resolved into an image that resembles an accurate root cross-section.

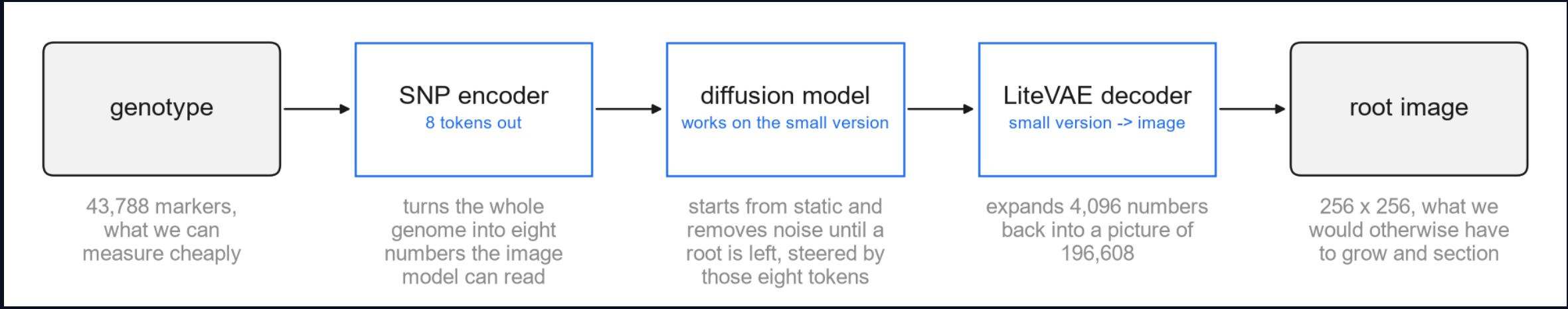
WHY DIFFUSION

It trains stably on a few hundred images, it takes conditioning naturally at every step, and a fixed starting noise always gives the same image, which is what makes controlled comparisons possible.

The diffusion model and where it sees DNA



The pipeline: read the genome, generate, decode

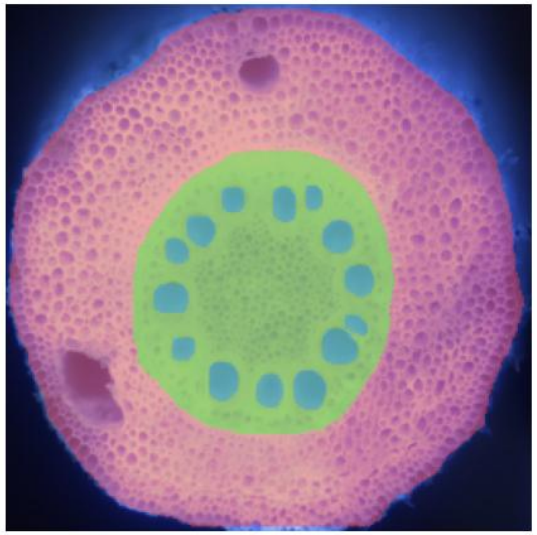


MEASURING WHAT IT DREW

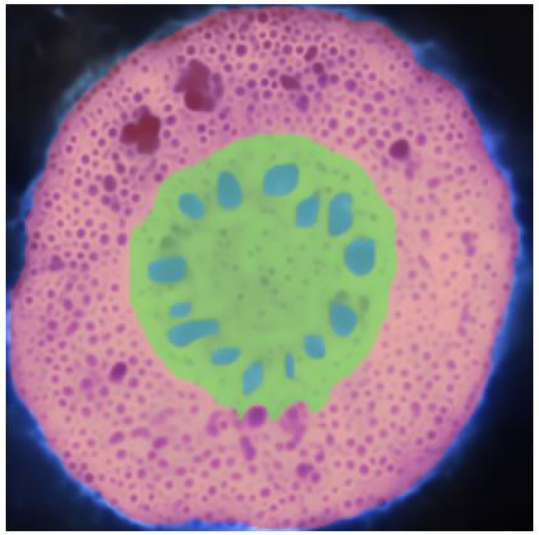
How can we compare real vs generated features?

MEMA042, seed 6 genotype MEMA042

Real image



Generated from SNPs only



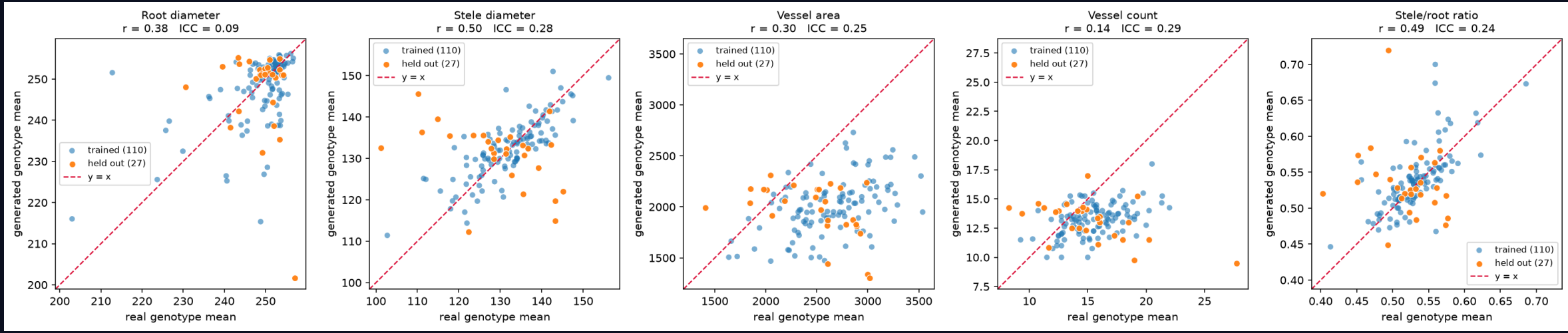
Trait	Real	Gen	Diff	%
Root diameter	251.5	252.1	+0.6	+0.2
Stele diameter	134.9	132.2	-2.7	-2.0
Vessel area	2505	2166	-339.0	-13.5
Vessel count	14	14	+0.0	+0.0
Stele/root ratio	0.536	0.524	-0.0	-2.2

red = root green = stele blue = vessel

A YOLOv8 segmentation model outlines the root, the stele and every vessel, then measures them: two diameters, vessel count, vessel area. The model runs on the original and on a generated image, which is what makes this a measurement rather than an impression.

DOES THE GENOTYPE COME THROUGH

Tracking trait correlation across entire dataset



ONE POINT PER LINE · BLUE = TRAINED ON · ORANGE = NEVER SEEN · DASHED LINE IS EXACT AGREEMENT

WHAT TRACKS

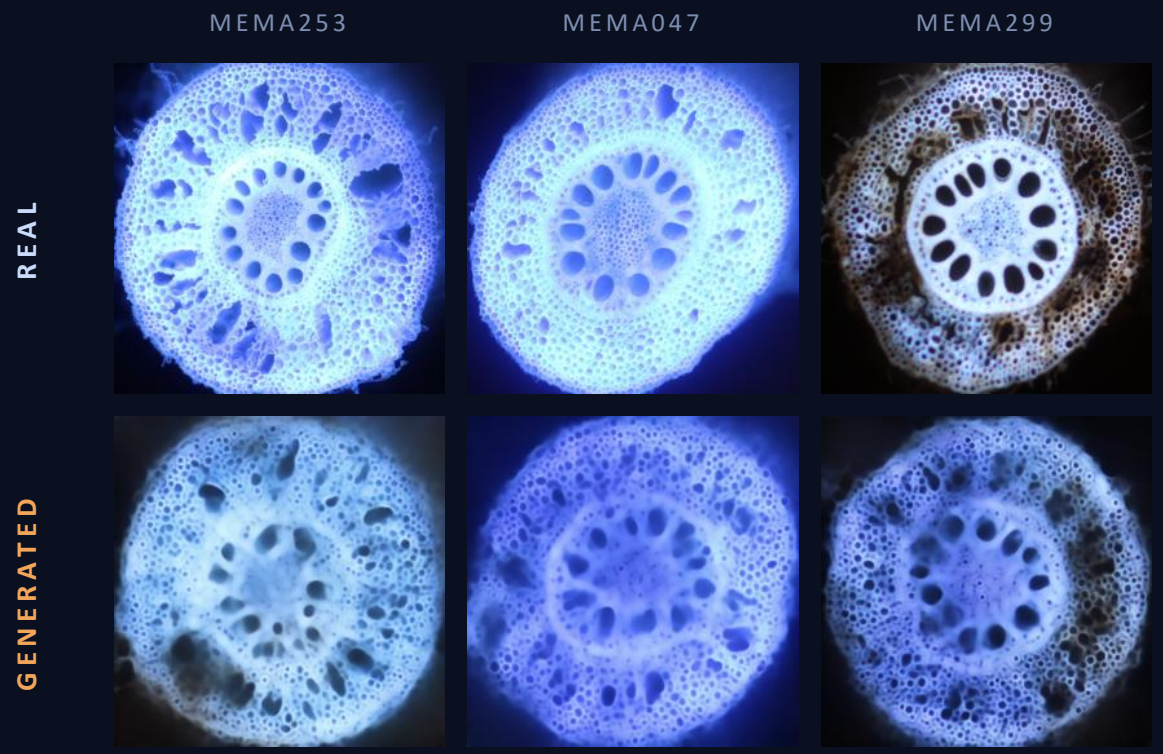
Trained lines (blue) track well: stele diameter r = 0.80, stele/root ratio 0.70. The 27 lines it never saw (orange) don't, so the r in each panel title is carried by the training set.

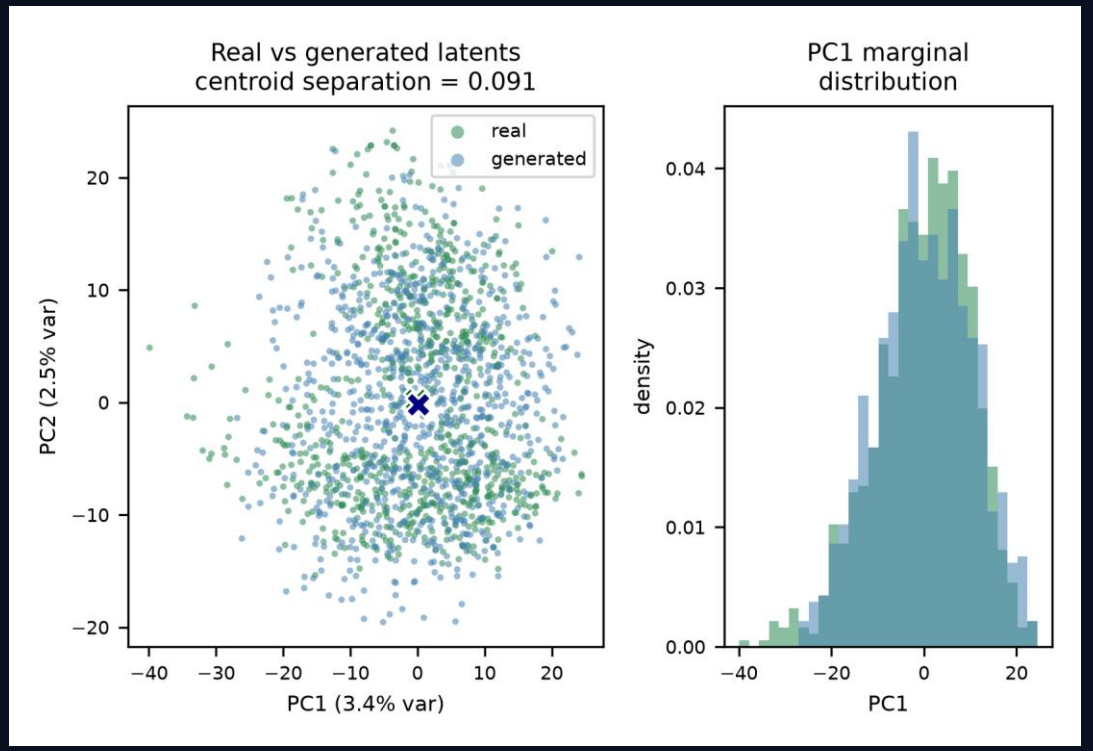
THE SEGMENTATION MODEL LIMITATION

The segmentation model in its current state struggles to reliably capture the vessel area and the count, often merging vessels together or missing certain vessels all together.

DO THEY LOOK REAL

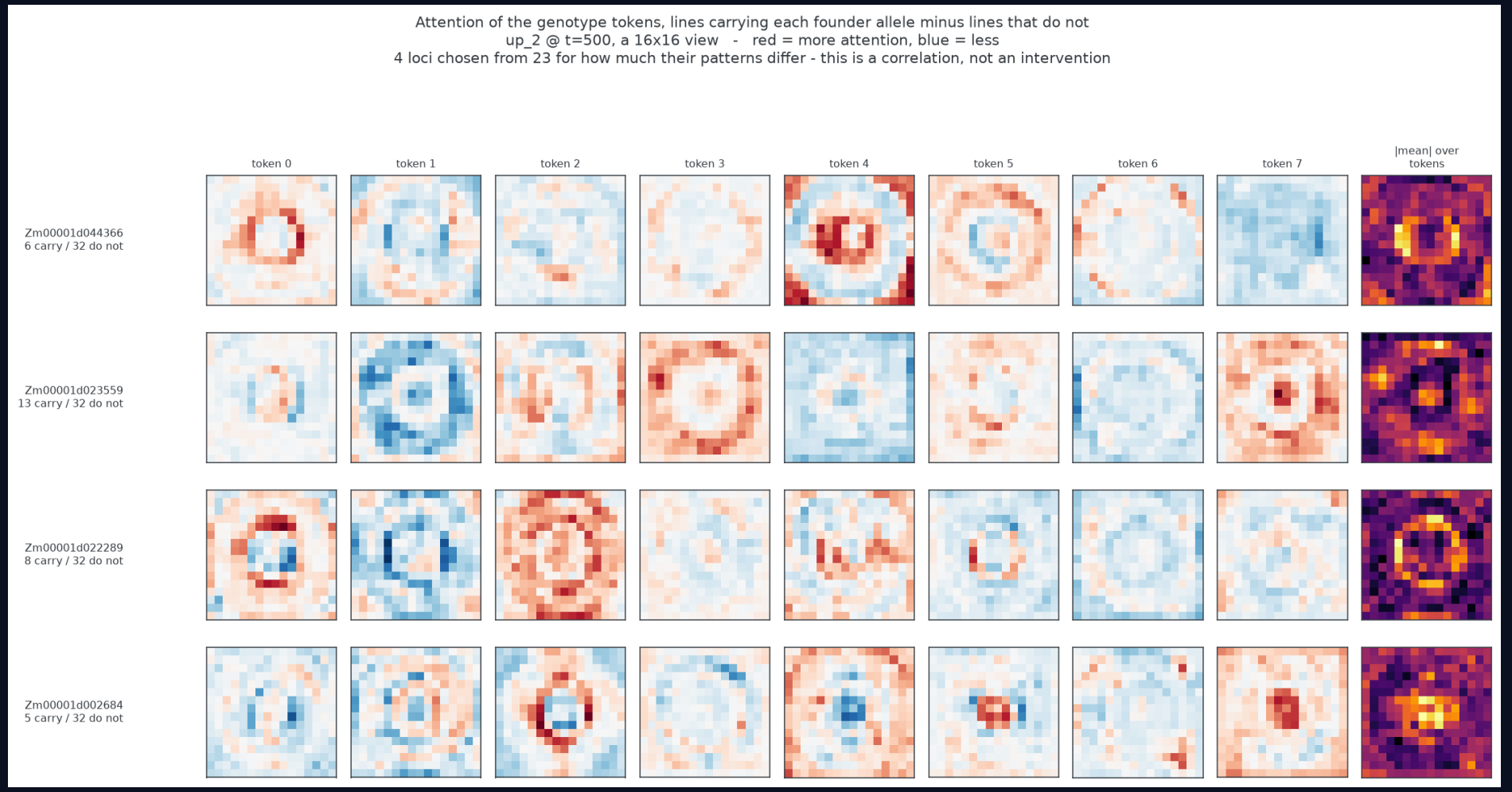
Generated roots look real, even in latent space





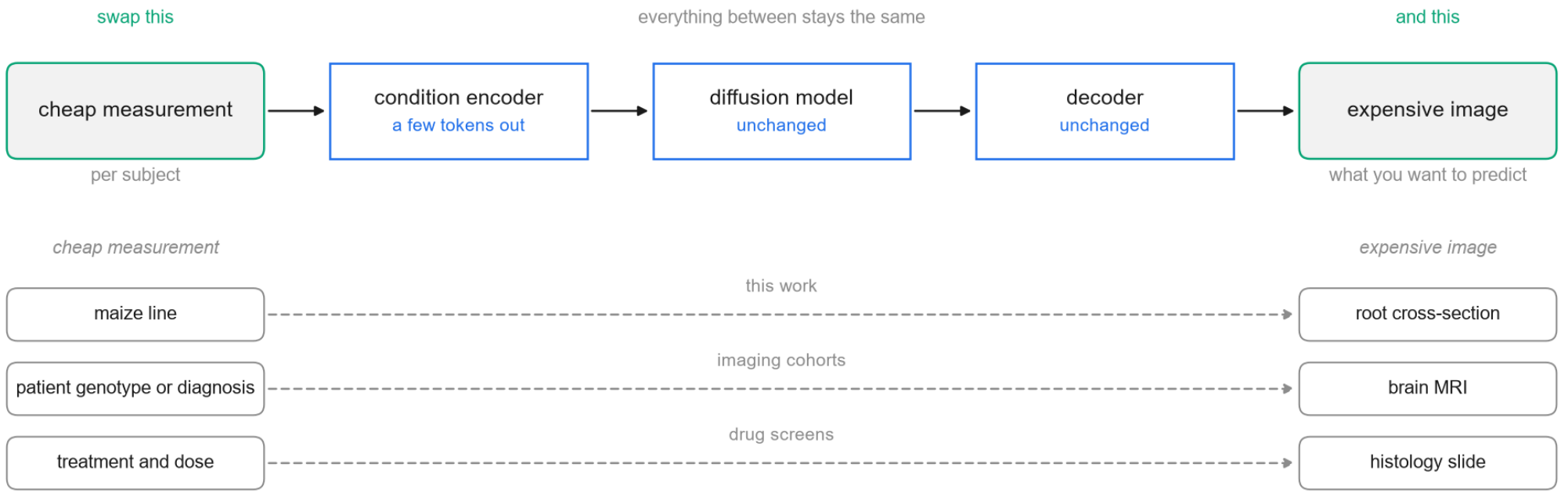
LATENT SPACE, ALL 866 IMAGES · GREEN = REAL · BLUE = GENERATED

What if we could map genes to phenes?

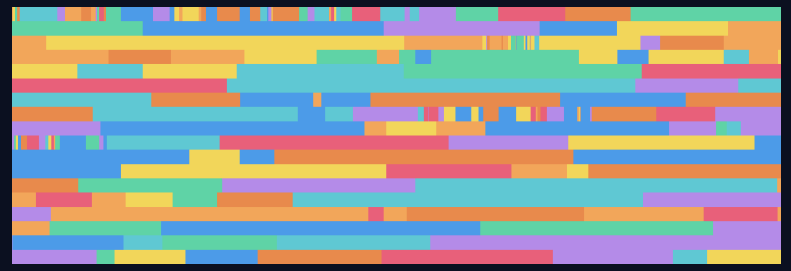


WHERE ELSE THIS FITS

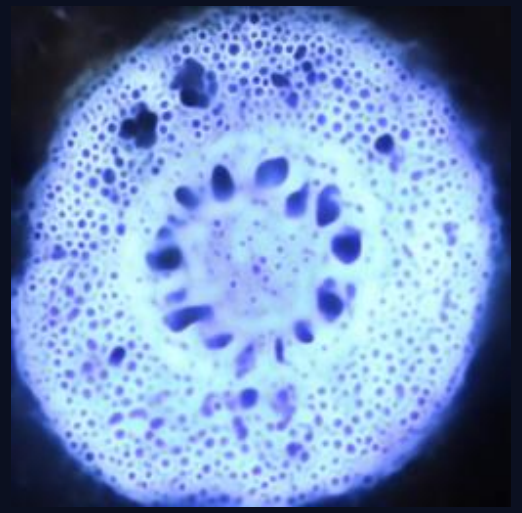
Swap the inputs: this works beyond roots and DNA



Genotype in. Anatomy out.



ONE GENOTYPE · 43,788 MARKERS



NO PLANT GROWN

Months of greenhouse, harvest and microscope.
Or a few seconds on a laptop.

WHAT COMES NEXT

Where this goes next

01 Improve the pipeline

Increase the accuracy of segmentation model, specifically with vessel segmentation, and perform pipeline on richer data.

02 Genomic prediction and gradient scoping

Compare against standard genomic prediction on the same traits, and trace gradients back to the markers that drive them.

03 New domains

The same three pieces on other paired data: other tissues, other crops, medical scans.

04 Your suggestions

What would you point this at?

THANK YOU

Acknowledgements



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University of Missouri



Jagdeep Sidhu

University of Missouri

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Melanie Perryman · Patrick Sydow · Sergio Perez Lemon

Questions?